

Microeconomic Theory: TA Session

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Bertrand competition

Suppose two firms produce identical products, and both have a marginal cost of 5. The firms simultaneously set price. The firm with the lowest price captures the entire market demand. If there is a tie in the lowest price, the two firms divide the market share equally. For simplicity, assume that market demand is fixed at 2.

1. Find the Bertrand equilibrium.

Answer: First, notice that no firm would set its price lower than the marginal cost of 5.

If $p_1 \geq p_2 > 5$: firm 1 will just lower price to $p_2 - \epsilon$ (with ϵ very small) and steal all the customers and profit. Same with $p_2 \geq p_1 > 5$.

So, the only possible equilibrium is $p_1 = p_2 = 5$: each firm sets price at the marginal cost, capturing half of the market share and earning zero profit.

Bertrand competition

Suppose two firms engage in Bertrand competition, and both have a MC of 5. For simplicity, assume that market demand is fixed at 2.

2. Now suppose prices can only change in the increment of one cent — you can set the price at \$5.45 or \$5.46, but not \$5.455. Is there an equilibrium? If so, find all equilibria. If not, explain.

Answer: There are three equilibria:

- ▶ (5,5) is still an equilibrium, just like in the original Bertrand problem where continuous pricing is allowed.
- ▶ (5.01, 5.01) is an equilibrium. Each firm's expected profit is \$0.01 ($= 0.01 \times 1$, since there are a total of 2 consumers). The only way for a firm to undercut is to lower price to \$5, but then this firm will earn zero profit, so no one has an incentive to deviate.
- ▶ (5.02, 5.02) is also an equilibrium. Each firm's expected profit is \$0.02 ($= 0.02 \times 1$). If a firm undercuts by lowering price to \$5.01, he captures the entire market but still earns a profit of \$0.02 ($= 0.01 \times 2$). So no one has an incentive to deviate.

(However, if demand is downward sloping, (5.02,5.02) is no longer an equilibrium)

Expected utility

For utility under uncertainty, we use the **expected utility function**

Suppose there are n possible cases with probability $p_1, p_2, \dots, p_n$ respectively (note that $p_1 + p_2 + \dots + p_n = 1$), and the consumption under each case is $c_1, c_2, \dots, c_n$. The utility function (with no uncertainty) is $u(c)$.

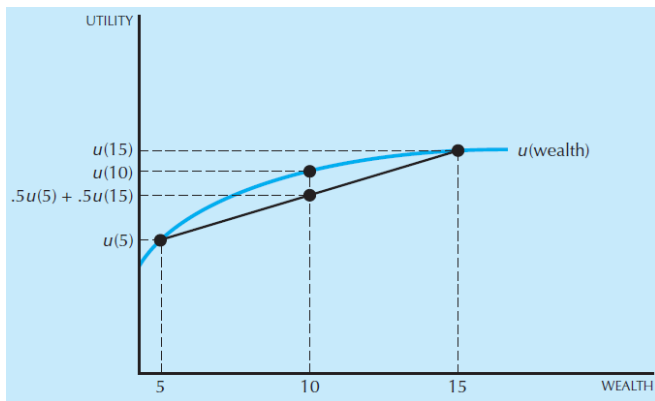
Then, the expected utility is:

$$p_1 u(c_1) + p_2 u(c_2) + \dots + p_n u(c_n)$$

Risk-averse, risk-neutral, and risk-loving

The shape of the utility function is associated with risk attitudes:

- Concave utility function — risk-averse



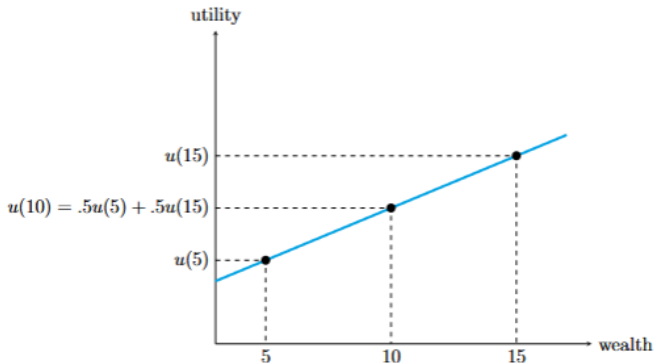
A concave utility function also exhibits diminishing marginal utility

- Intuition: Diminishing marginal utility $\Rightarrow$ he hates losing \$5 more than he likes gaining \$5 $\Rightarrow$ Risk aversion

Risk-averse, risk-neutral, and risk-loving

The shape of the utility function is associated with risk attitudes:

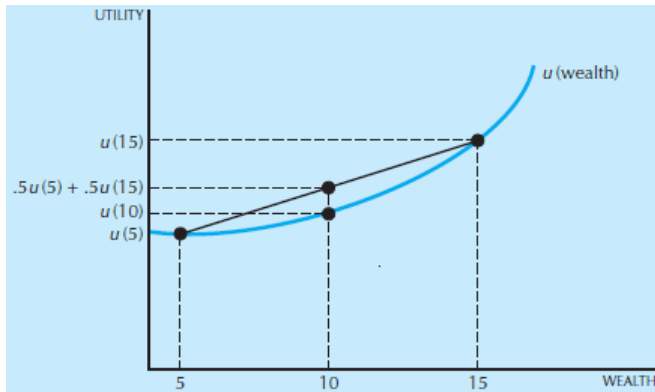
- Linear utility function — risk-neutral



Risk-averse, risk-neutral, and risk-loving

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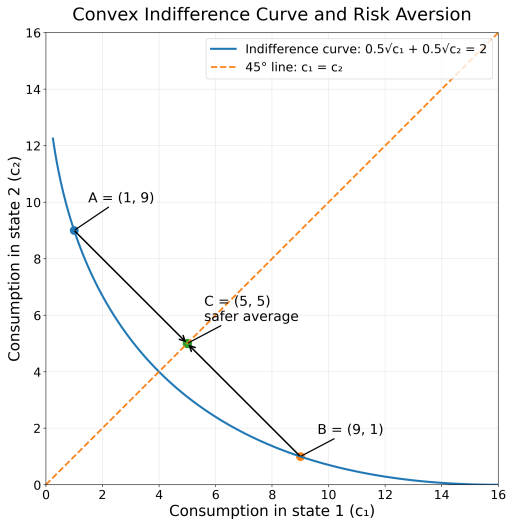
- Convex utility function — risk-loving



Risk-averse, risk-neutral, and risk-loving

The shape of indifference curves is associated with risk attitudes:

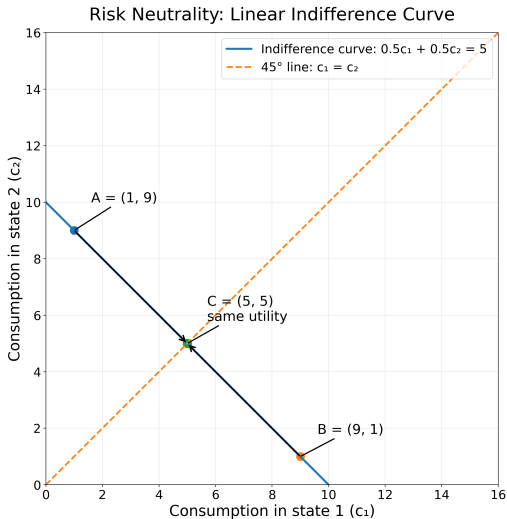
- Convex indifference curves — risk-averse (consumer prefers C over A or B)



Risk-averse, risk-neutral, and risk-loving

The shape of indifference curves is associated with risk attitudes:

- ▶ Linear indifference curves — risk-neutral (consumer is indifferent between C, A or B)



Risk-averse, risk-neutral, and risk-loving

The shape of indifference curves is associated with risk attitudes:

- Concave indifference curves — risk-loving (consumer prefers A or B to C)

